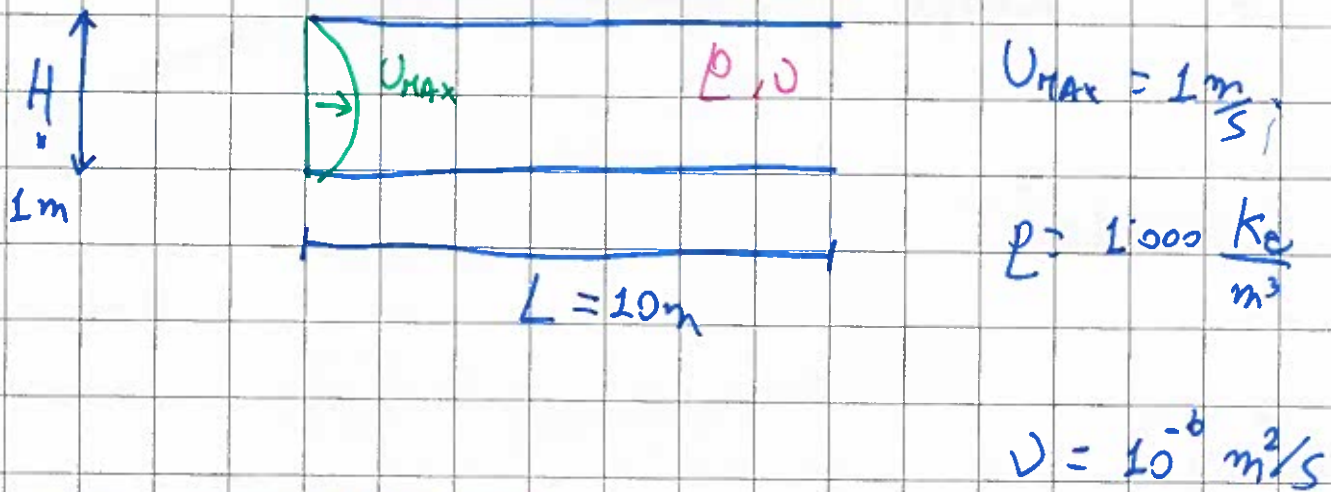




$$H^{LB} = 10$$

$$S_L = \frac{H}{H^{LB}} = \frac{1\text{m}}{10} = 0.1\text{m} = \Delta x$$

$$L^{LB} = \frac{L}{S_L} = \frac{10\text{m}}{0.1\text{m}} = 100$$

$$Ma = \frac{U_{\text{max}}^{LB}}{C_s} = 0.01\sqrt{3}$$

$\textcircled{C_s} \downarrow$
 $= \frac{1}{\sqrt{3}}$

$$U_{\text{max}}^{LB} = 0.01$$

$$S_v = \frac{U_{\text{max}}}{U_{\text{max}}^{LB}} = \frac{1\text{m/s}}{0.01} = 100\text{m/s}$$

$$S_t = \frac{S_L}{S_v} = \frac{0.1\text{m}}{100\text{m/s}} = 10^{-3}\text{s} = \Delta t$$

$$\rho^{LB} = 1, \quad S_\rho = \frac{\rho}{\rho^{LB}} = 100, \frac{\text{kg}}{\text{m}^3}$$

$$S_v = \frac{S_L^2}{S_t} = \frac{(0.1 \text{ m})^2}{0.001 \text{ s}} = \frac{0.01 \text{ m}^2}{0.001 \text{ s}} = 10 \frac{\text{m}^2}{\text{s}}$$

$$v^{LB} = \frac{v}{S_v} = 10^{-7}$$

$$\frac{1}{\tau} (8^9 - 8)$$

↓

RELAXATION TIME $\rightarrow v^{LB} = \left(\tau - \frac{1}{2} \right) C_s^2 \rightarrow \tau = \frac{10^{-7}}{C_s^2} + \frac{1}{2} =$

$$= 0.5 + 3 \cdot 10^{-7}$$